C PROGRAMMING — ASSIGNMENT 1

Task 2: Choose the Data Types

Marks: 4 | Total assignment: 50 marks

Type choices and ranges

Use uint16_t for a meter register only when the implementation provides the exact-width type; the assignment's typedef unsigned short meter_t expects 2 bytes and USHRT_MAX == 65535 on the target system. Use int for small bounded values, uint32_t for units, and int64_t/long long for monetary totals. Use double only for measurements or calculations where approximate floating point is acceptable, not for accounting totals.

The exact sizes are implementation-dependent. Run the supplied program on the target machine; do not assume every C implementation uses the same sizeof result. limits.h defines integer limits; float.h defines floating-point characteristics such as FLT_DIG, DBL_DIG and FLT_EPSILON.

Money and floating-point example

A decimal fraction such as 0.1 is not exactly representable in binary floating point. Repeated additions can accumulate a small representation error. For example, repeated addition of 0.1f can differ from the mathematically expected value of n/10. For billing, represent Rs 162.50 as 16250 paise; addition and subtraction of integer paise remain exact as long as the integer type does not overflow.

Rollover

For a 16-bit unsigned register, subtraction wraps modulo 65536. Therefore (uint16_t)(current - previous) gives 330 for 1200→1530 and 256 for 65400→120. The second result is not negative: unsigned arithmetic wraps modulo 2^16, so 120 - 65400 ≡ 256 (mod 65536). This assumes at most one rollover between readings.

Example size/range program

Compile and run the code below to collect your machine's actual sizeof values and limits. The output shown in the assignment expects unsigned short to be 2 bytes and maximum 65535.

C Source Code / Implementation

```
#include <stdio.h>
#include <limits.h>
#include <float.h>
#include <stdint.h>

typedef unsigned short meter_t;

static unsigned int consumption(meter_t previous, meter_t current) {
    return (meter_t)(current - previous);
}

int main(void) {
    printf("sizeof(char)=%zu, sizeof(short)=%zu, sizeof(int)=%zu\n",
           sizeof(char), sizeof(short), sizeof(int));
    printf("sizeof(long)=%zu, sizeof(long long)=%zu, sizeof(float)=%zu, sizeof(double)=%zu\n",
           sizeof(long), sizeof(long long), sizeof(float), sizeof(double));
    printf("USHRT_MAX=%u, INT_MAX=%d, LONG_MAX=%ld, LLONG_MAX=%lld\n",
           (unsigned)USHRT_MAX, INT_MAX, LONG_MAX, LLONG_MAX);
    printf("FLT_DIG=%d, DBL_DIG=%d\n", FLT_DIG, DBL_DIG);
    printf("sizeof(meter_t) = %zu bytes, maximum reading = %u\n",
           sizeof(meter_t), (unsigned)USHRT_MAX);
    printf("Consumption: %u units\n", consumption(1200, 1530));
    printf("Consumption: %u units\n", consumption(65400, 120));
    return 0;
}
```

Submission note

Compile and run the code on your own system, save actual terminal output/screenshots, and include them in your report. Review the code and be prepared to explain it during the viva. Any machine-specific measurements must come from your own execution.